Pertussis and the CMI-PB project

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##Background Pertussis (Whooping cough) is a common lung infection caused by the bacteria B. Pertussis. It can infect everyone but is most deadly for infants (under 1 year).

CDC tracking data

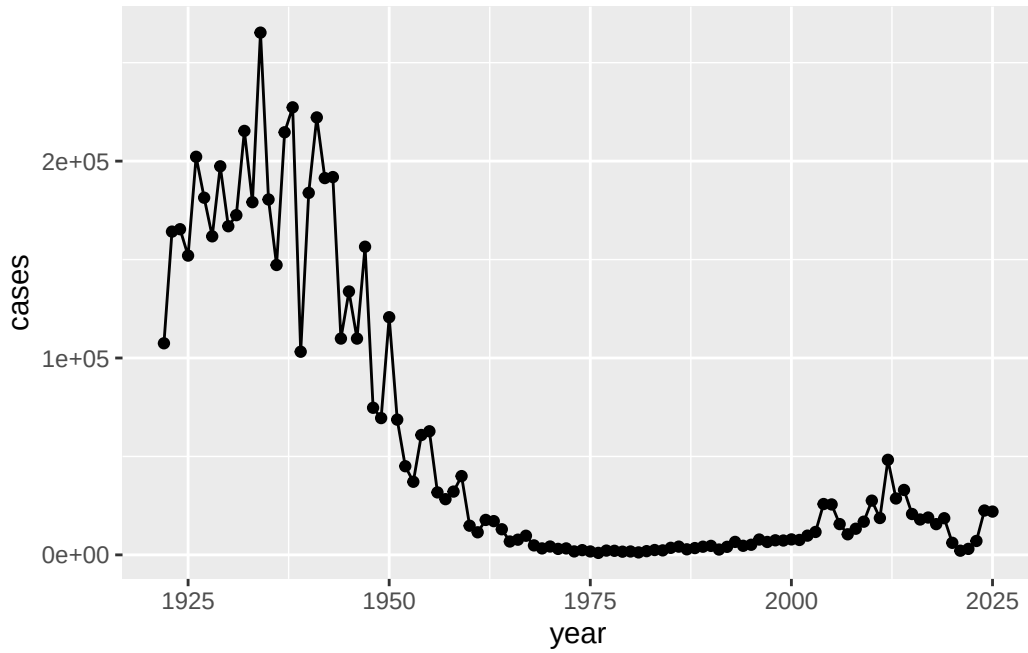
the CDC tracks the number of pertussis cases:

Q1. I want a plot of year vs cases

```
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2
```

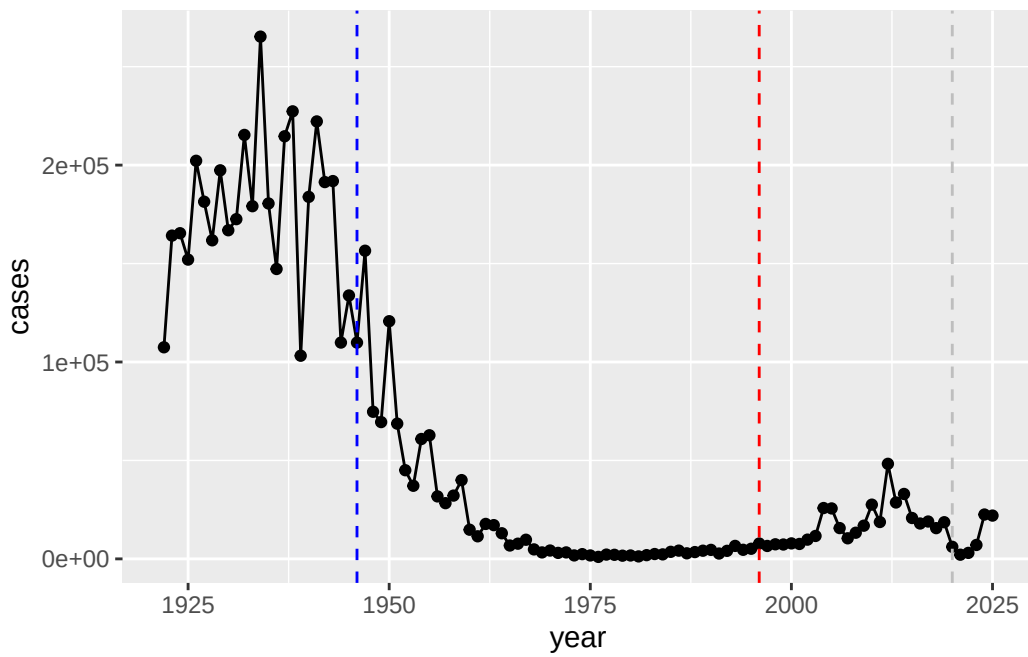
```
ggplot(cdc) +
  aes(year, cases) +
geom_point() +
geom_line()
```



Q2. Add annotation line for the major milestone of wP vaccination roll-out (1946) and switch to the aP vaccine9 (1996). >Q2. Using the ggplot geom_vline() function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
library(ggplot2)

ggplot(cdc) +
  aes(year, cases) +
geom_point() +
geom_line() +
geom_vline(xintercept = 1946, col="blue", lty=2) +
geom_vline(xintercept = 1996, col="red", lty=2) +
geom_vline(xintercept = 2020, col="gray", lty=2)
```



Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

After the aP vaccine was introduced, pertussis cases started increasing again. This could be because immunity from the aP vaccine may wear off faster, better PCR testing detects more cases, and the bacteria may have evolved to escape vaccine immunity.

##Exploring CMI-PB data the CMI- PB project < https://www.cmi-pb.org/ > The mission of CMI-PB is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of Pertussis booster vaccination.

Basically, make available a large datasets on the immune response to Pertussis. They use a "booster" vaccination as a proxy for Pertussis infection.

They make their data available as JSON format API. We can read this into R with the `rea_JSON()` function from the **jsonlite** package:

```
library(jsonlite)
subject <- read_json("https://www.cmi-pb.org/api/v5_1/subject", simplifyVector = TRUE)

head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
```

```
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

Q4. How many aP and wP individuals are there in this `subject` table?

```r
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset? How many male/female are there?

```r
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

Q. What is the break down of `biological_sex` gender and `race`?

Q. Is this representative of the US population? Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc…)?

```r
table(subject$race, subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
```

```
  Black or African American                       2    3
  More Than One Race                             15    4
  Native Hawaiian or Other Pacific Islander       1    1
  Unknown or Not Reported                        14    7
  White                                          48   32
```

We can read more table from the CMI-PB database

```r
specimen <- read_json("https://www.cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE )
ab_titer <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRU

head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```r
head(ab_titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
```

```
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To make sense of all this data we need to "join" (a.k.a "merge" or "link") all these tables together. Only then will you know that a given Ab measurement (from the `ab_titer` table) was collected on a certain data (from the `specimen` table) from a certain wP or aP subject (from the `subject` table).

We can use **dplyer** as the `*_join()` family of functions to do this.

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
```

```
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           11                            14         Blood
6                           32                            30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

Let's do one more `inner_join()` to join the `ab_titer` with all our `meta` data.

```r
abdata <- inner_join(ab_titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
6 IU/ML                 2.816431          1          wP         Female
```

```
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Q11. How many different Ab "isotype" values are in this dataset?

```r
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

Q. How many different "antigen" values are measured?

```r
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA 
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318 
    PD1     PRN      PT     PTM   Total      TT 
   1970    6712    6712    1970     788    6318 
```

Q12. What are the different $dataset values in abdata and what do you notice about the number of rows for the most "recent" dataset?

```r
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
       31520         8085         7301        15050
```

Let's focus on IgG isotype:

```r
igg <- abdata |>
          filter (isotype=="IgG")

head(igg)
```

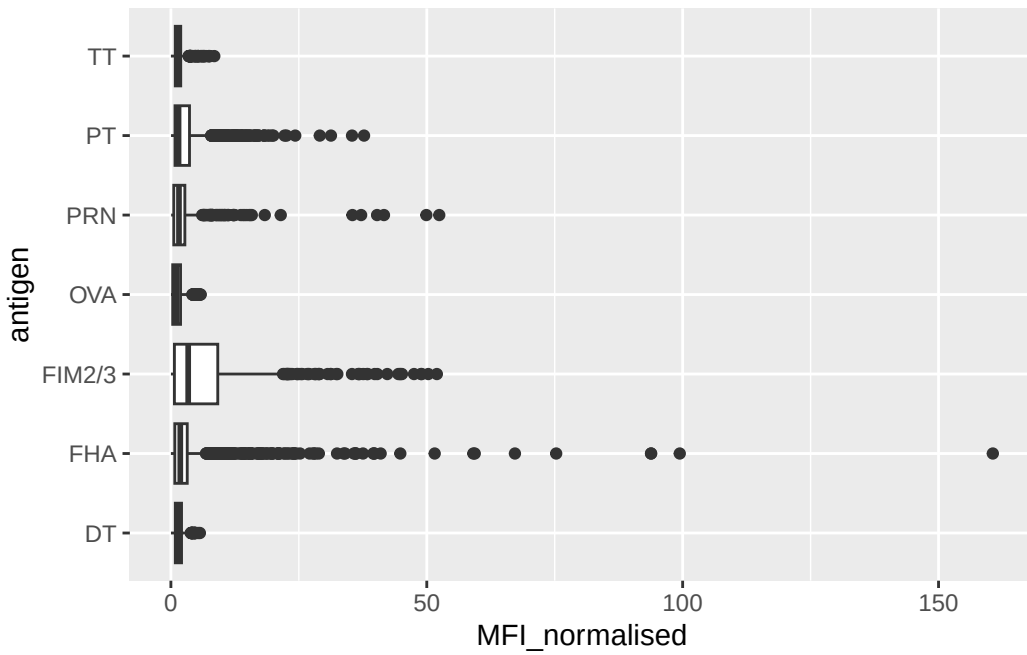
```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 IU/ML                 0.530000          1          wP         Female
2 IU/ML                 6.205949          1          wP         Female
3 IU/ML                 4.679535          1          wP         Female
4 IU/ML                 0.530000          3          wP         Female
5 IU/ML                 6.205949          3          wP         Female
6 IU/ML                 4.679535          3          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
```

```
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

Make a plot of `MFI-normalised` values for all `antigen` values.

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```



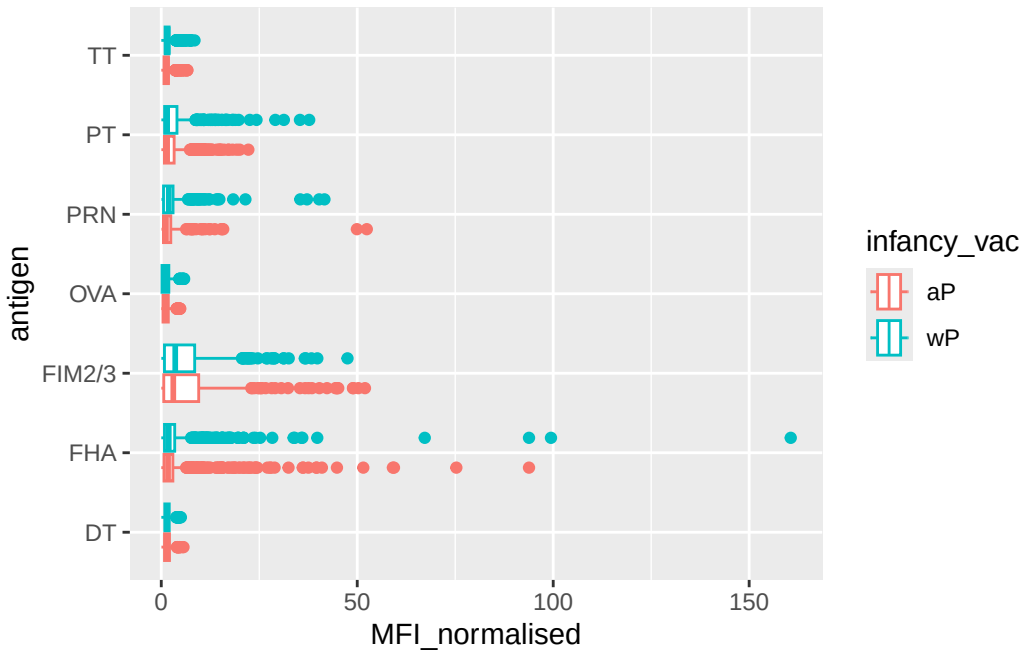
Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

PT, PRN, FHA, and FIM2/3 show the biggest changes in IgG levels over time because their boxplots shift more between visits. The other antigens stay mostly low or similar across visits, so they don't show much change.

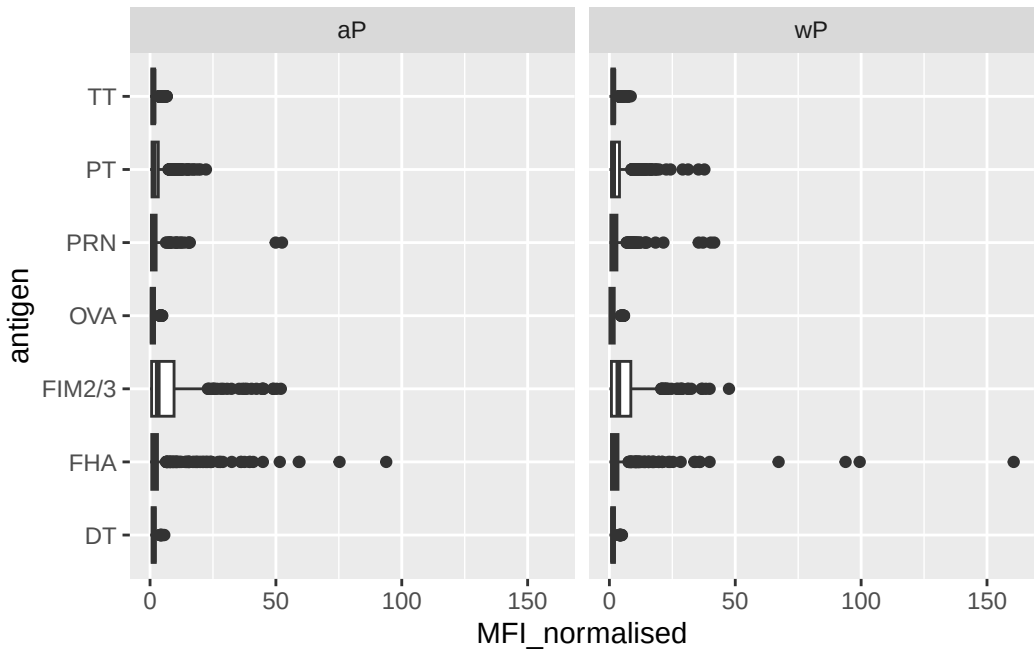
The antigen "PT", "FIM2/3" and "FHA" appear to have the widest range of values.

Q. Is there a differnce for these responses between aP and wP individuals?

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot()
```

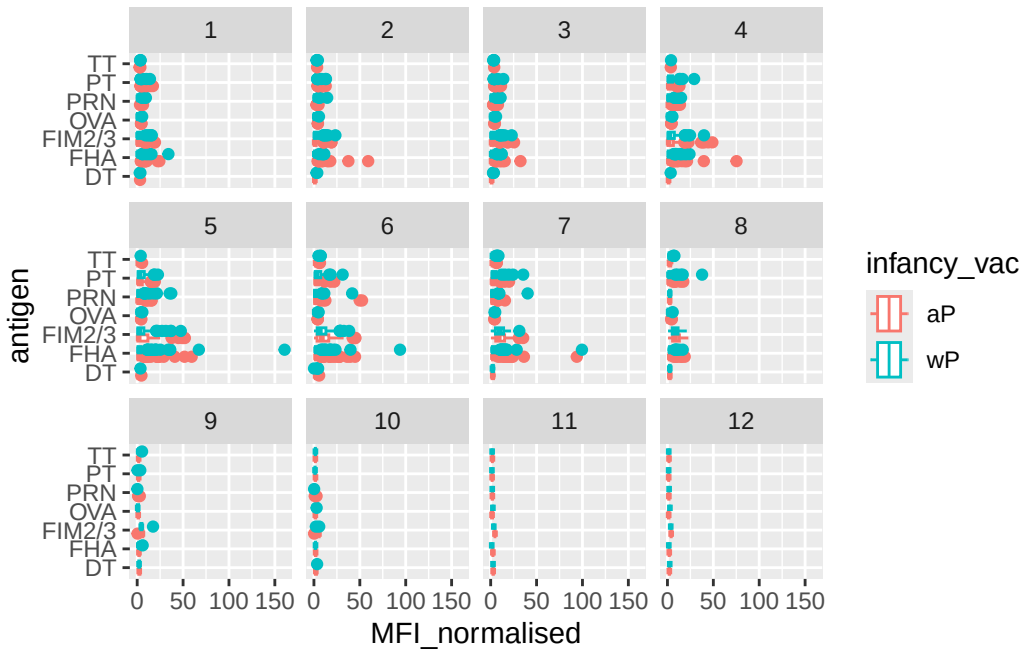



```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
facet_wrap(~infancy_vac)
```



Q. Is there a difference with time (i.e. before booster shot vs after booster shot)?

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot() +
facet_wrap(~visit)
```

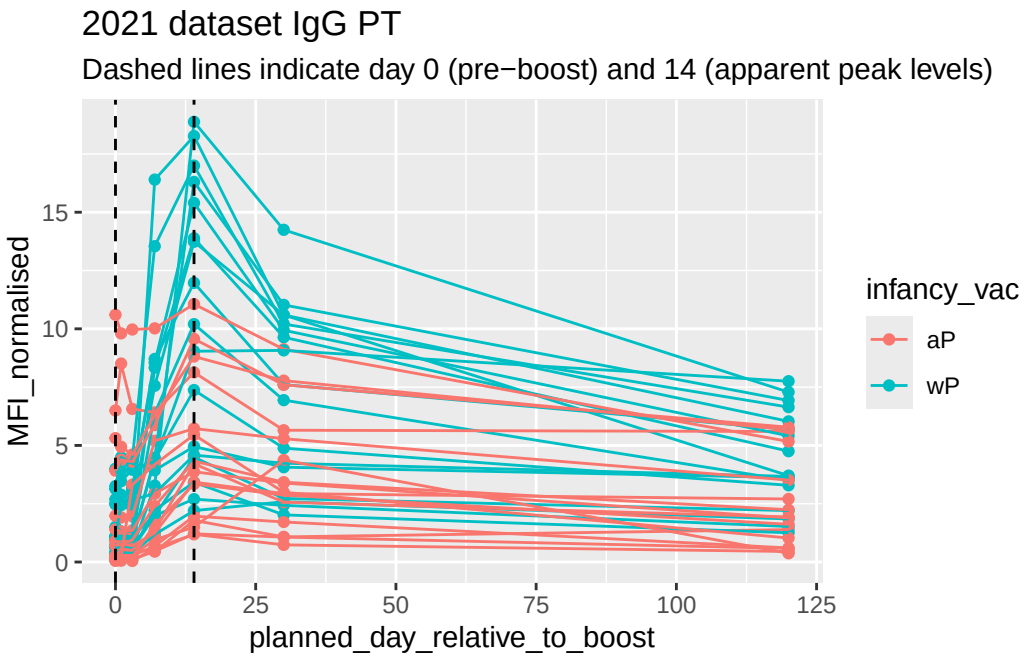


Lets finish this section by looking at the 2021 dataset IgG PT antigen levels time-course:

```
## filter to 2021 dataset

abdata.21 <- abdata |>
  filter(dataset == "2021_dataset",
         isotype =="IgG",
         antigen =="PT")

ggplot(abdata.21) +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
  labs(title="2021 dataset IgG PT",
       subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

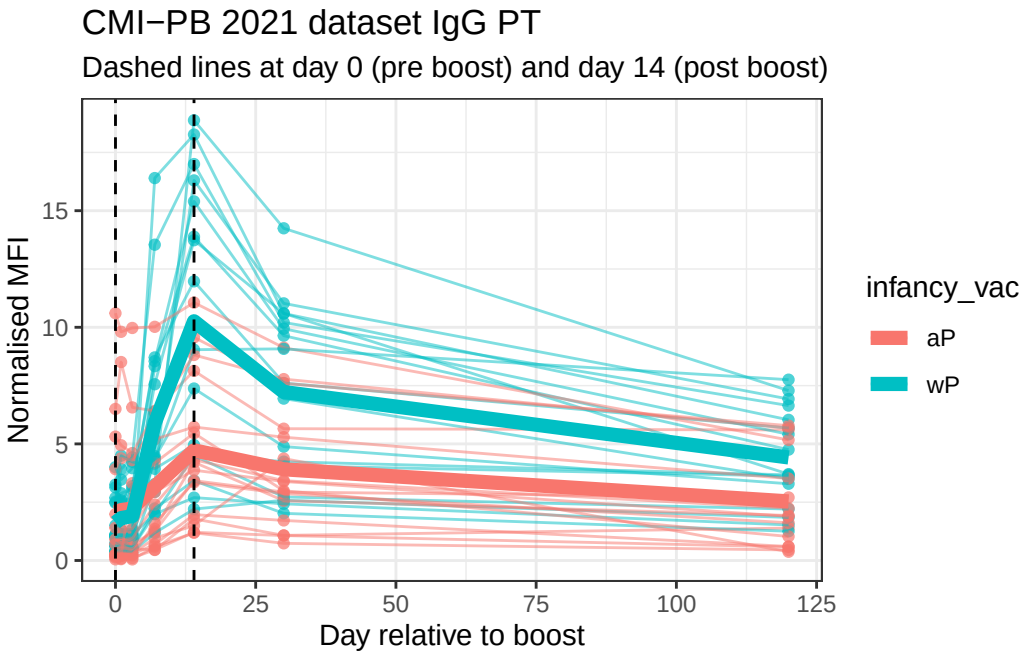



```
abdata.21 <- abdata |>
  filter(dataset == "2021_dataset",
         isotype == "IgG",
         antigen == "PT")

ggplot(abdata.21) +
  aes(x = planned_day_relative_to_boost,
      y = MFI_normalised,
      col = infancy_vac,
      group = subject_id) +
  geom_point(alpha = 0.7) +
  geom_line(alpha = 0.5) +

  # thick trend lines
  stat_summary(aes(group = infancy_vac),
               fun = mean,
               geom = "line",
               linewidth = 2.5) +

  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = 14, linetype = "dashed") +

  labs(
```

```
    title = "CMI-PB 2021 dataset IgG PT",
    subtitle = "Dashed lines at day 0 (pre boost) and day 14 (post boost)",
    x = "Day relative to boost",
    y = "Normalised MFI"
  ) +
  theme_bw()
```